CASE STUDY · PHARMA WAREHOUSE MANAGEMENT SYSTEM · 2026

Digitising Dispatch Management for a Pharma WMS

Designing a connected dispatch workflow from shipment readiness to delivery outcome helped **save an estimated 5+ hrs/week in manual effort,** based on stakeholder inputs.

Role	Individual Product Designer
Domain	Pharma Logistics · Warehouse Ops
Scope	Discovery · IA · UX · UI · Prototyping

01 Overview

From shipment readiness to delivery outcome

I designed a connected dispatch workflow for a pharmaceutical WMS, covering the shipment lifecycle from dispatch readiness through delivery outcome. The workflow brings shipment visibility, load assignment, validation, tracking, POD, and exception handling into one system.

Dispatch Workbench → Load Management → Release → Tracking → POD → Delivery Outcome → Exception / Redelivery

02 Problem

Dispatch involved multiple operational steps and decisions, but the process depended heavily on manual coordination. The opportunity was to bring the complete dispatch lifecycle into one digital workflow and make critical decisions easier to track, validate and audit.

> **The opportunity**
> Bring the dispatch lifecycle into one digital workflow while keeping decisions clear, controlled, and traceable.

03 Goals

Turn dispatch into one controlled workflow.

01
Visibility
Give dispatchers one place to see shipment status and exceptions.

02
Guided decisions
Use shipment and compliance information to prevent incorrect actions.

03
Traceability
Keep dispatch, POD, and exception decisions linked to the shipment.

04
Less manual coordination
Move dispatch activities into the WMS instead of relying on scattered manual processes.

04 Research

I mapped the dispatch lifecycle and its decision points.

I worked from the BRD and mapped the workflow across shipment readiness, load creation, resource assignment, validation, release, delivery, POD, and exceptions. Then identified where the system needed to guide or block a decision instead of simply displaying information.

01
What is ready?
→ Dispatch Workbench

02
How do we prepare it?
→ Load Management

03
Can it safely leave?
→ Validation / Release

04
What happened?
→ Tracking / POD

05
What do we do next?
→ Exception handling

> **Key finding**
> The system needed to do more than show shipment data. It needed to help dispatchers make the right operational decision. This became the basis for how I structured the workflow: show what needs attention, guide the next decision, and prevent unsafe actions.

05 Solution & Tradeoffs

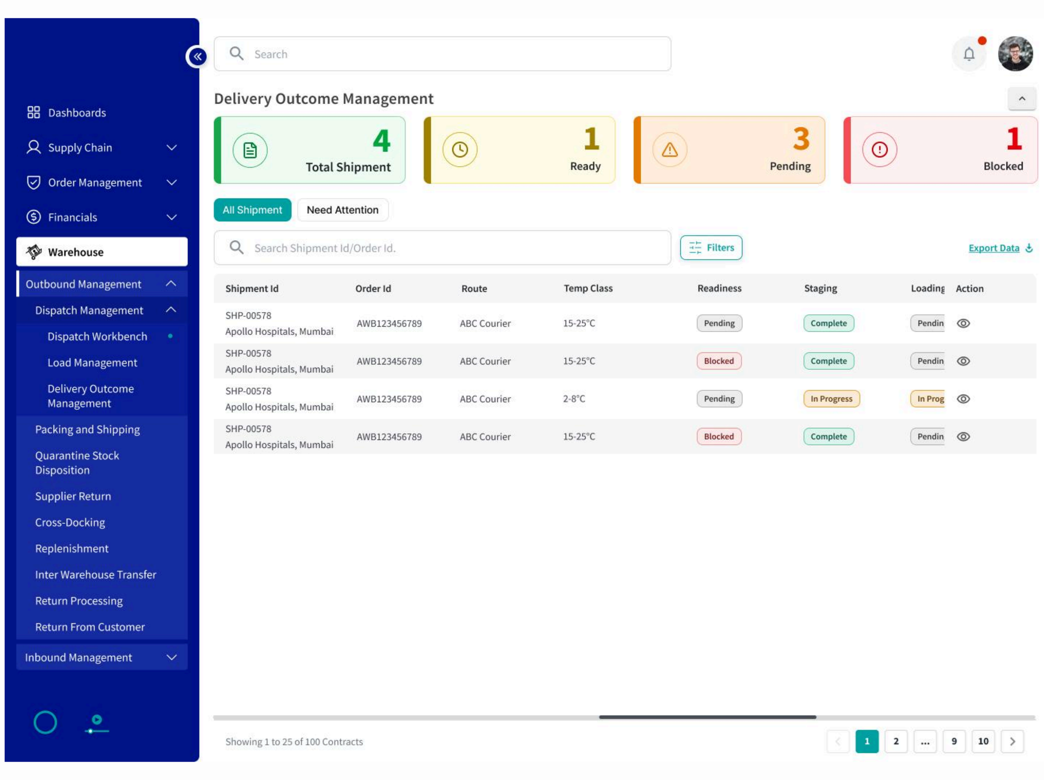
1. See what needs attention

Problem
Dispatcher needs to know what is ready, blocked and where each shipment is in the process.

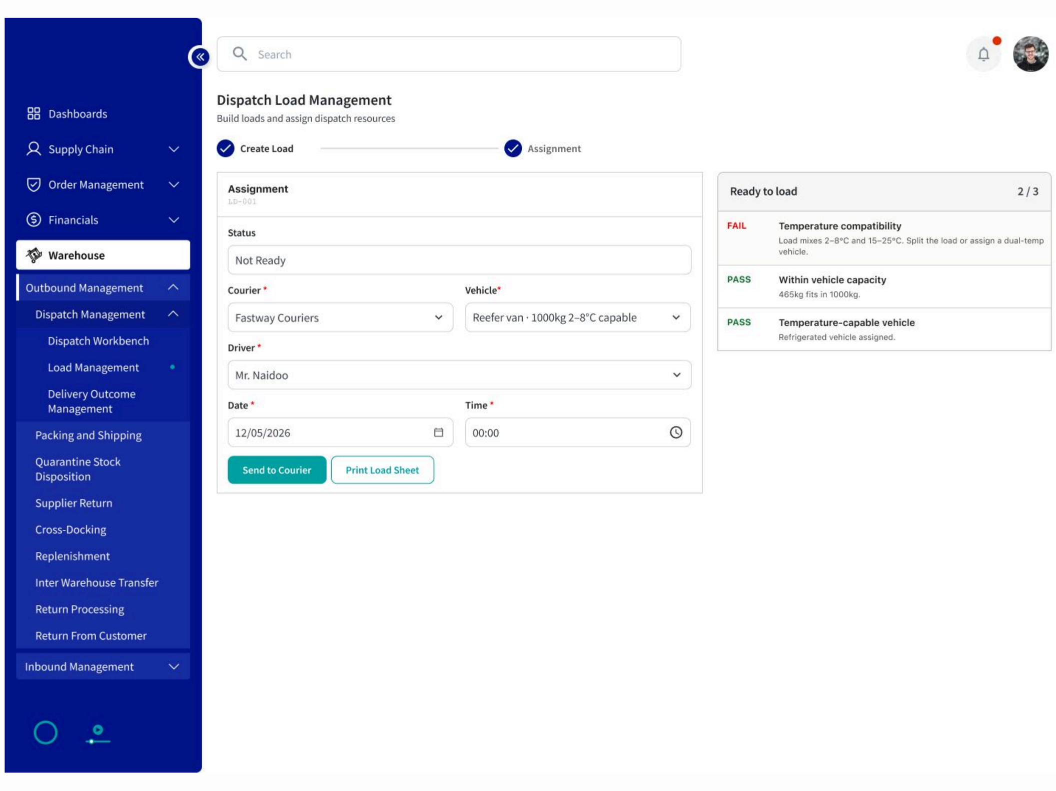
Decision
Bring the shipment lifecycle into one operational view.

Solution
Dispatch Workbench - I surfaced readiness, staging, loading, compliance and release status together so the dispatcher could identify what needs attention without opening every shipment.

Trade-off
More information vs. quick scanning
Showing every shipment detail in the workbench could make the table harder to scan.



Instead of checking different stages separately, the dispatcher can see the shipment state at a glance.



2. I moved an important operational check into the assignment step.

Problem
A shipment may have specific requirements, such as temperature range and vehicle capacity. Assigning the wrong vehicle can create a compliance or delivery risk.

Decision
Validate vehicle suitability while the dispatcher is assigning the load, rather than after the assignment.

Solution
I connected shipment requirements with vehicle checks so the dispatcher can see temperature compatibility and capacity before sending the shipment forward.

Trade Off
Flexible assignment vs. operational safety
Allowing any vehicle gives the dispatcher more flexibility, but it can lead to an incompatible assignment.

3. Release only when checks are complete

Problem
A shipment should not move to dispatch when required operational checks are incomplete.

Decision
Make release dependent on the required checks instead of treating it as a standalone action.

Solution
Release validation - I used shipment readiness, staging, loading, compliance, and blocking exceptions to determine whether a shipment can move forward.

Staging complete
↓
Loading complete
↓
Required checks passed
↓
Ready to release

- Delivery outcome
- Receiver details
- Carton count
- Signature
- Delivery photo
- Delivery notes

POD Confidence
Low · Medium · High
POD confidence is based on the information captured during delivery. If the required information is complete and accurate, confidence increases.

4. Connect delivery evidence to the shipment

Problem
After dispatch, the dispatcher needs to know what happened at delivery and whether the delivery evidence is complete

Decision
Connect the driver's delivery inputs with the shipment's delivery outcome.

Solution
POD & Delivery Outcome - The driver captures the delivery outcome, receiver details, carton count, signature, delivery image, and notes. These inputs contribute to POD confidence and help the dispatcher understand the delivery status.

5. Handle cold-chain breaches differently

Problem
A cold-chain breach is different from a normal delivery failure because the shipment may no longer be safe to send to the patient.

Product decision models I explored

Option 01 — REJECTED
Warning-based guidance
- × Easy to ignore under pressure
- × Unsafe path stays available
- × Compliance still depends on memory

Option 02 — REJECTED
Hard lock on breach
- × Too restrictive no recovery path
- × Stops legitimate scenarios
- × Breaks real operational work

Option 03 — CHOSEN
Guided restriction
- ✓ Unsafe actions removed
- ✓ Safe actions kept available
- ✓ QA folded into the flow
- ✓ Auditable, and operationally scalable

Trade-Offs I Accepted

Safety over speed
The extra check adds a few seconds but stops an unsafe box from leaving the building.
Safety Won

Compliance over cost
Returning and writing off the stock costs money cheaper than harming a patient.
Worth It

Flexible vs strict
QA can override instead of blocking forever. The cost: a good box waits for disposition.
Controlled Override

06 Impact

Estimated manual effort saved 5+ hrs/week

Before
- Manual coordination
- Scattered information
- Decision depends on individual knowledge
- Difficult to trace the full process

After
- One connected dispatch workflow
- Shipment status visible
- System-guided validation
- Structured POD
- Exception handling
- Audit trail